Systems Analysis and Control

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Lecture 14: Root Locus
Asymptotic Center, Break Points, Departure Angles

Overview

In this Lecture, you will learn:

Asymptotic Center

- Where do the asymptotes meet?

Where does the Root Locus Leave the Real Axis

- Break Points

What is the effect of small gain?

- Departure Angles from Complex Poles

What is the behaviour at high gain?

- Arrival Angles at Complex Zeroes

Root Locus

Review of Asymptotes

Pole locations change at high gain.

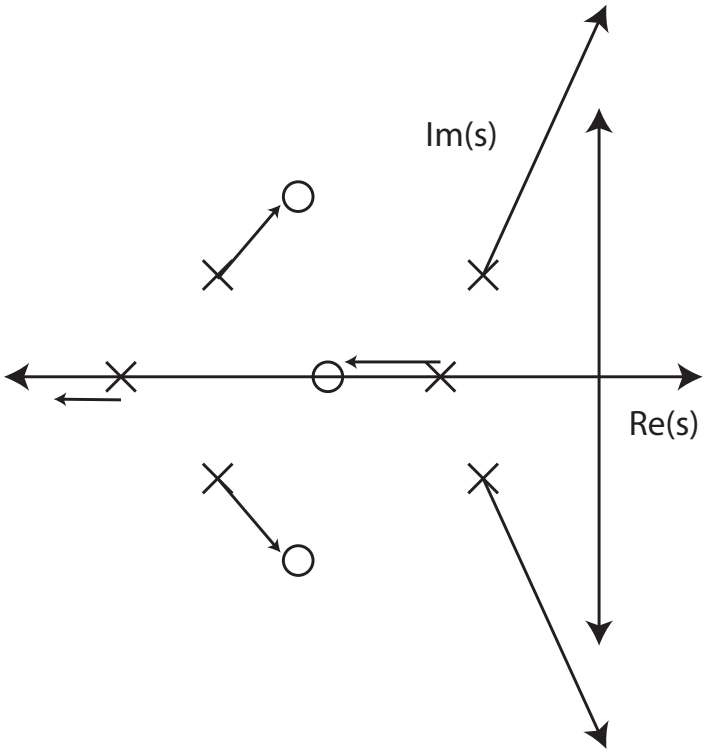
- Some poles stay small
- Some poles get large
  - Asymptotes depend on relative number of poles and zeros.

Small poles go to zeros.

Big poles leave on asymptotes:

Cases:

- $n - m = 0$ - No Asymptotes
- $n - m = 1$ - Asymptote at $180°$
- $n - m = 2$ - Asymptotes at $\pm 90°$
- $n - m = 3$ - Asymptotes at $180°$, $\pm 60°$
- $n - m = 4$ - Asymptotes at $\pm 45°$ and $\pm 135°$



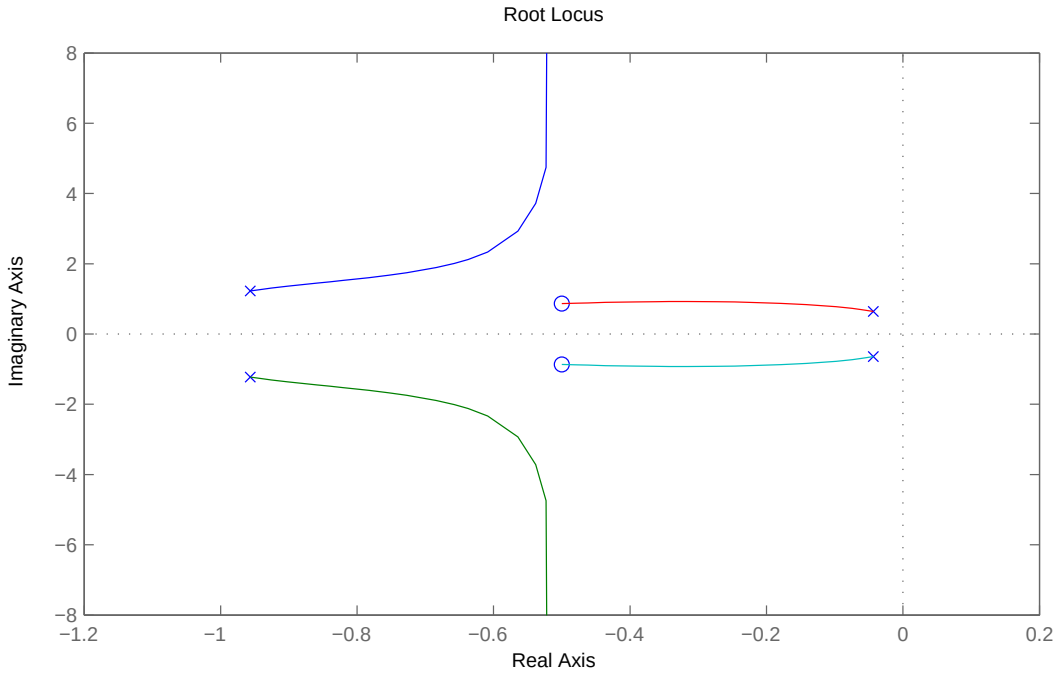
Root Locus: Where do asymptotes meet?

90° Asymptotes

Recall the suspension system:
$G(s) = \frac{s^2+s+1}{s^4+2s^3+3s^2+1s+1}$.

Count: 2 zeros, 4 poles.

$$n - m = 2$$



$$\angle_\infty = -90^\circ, -270^\circ$$

2 vertical asymptotes at 90° and 270°.

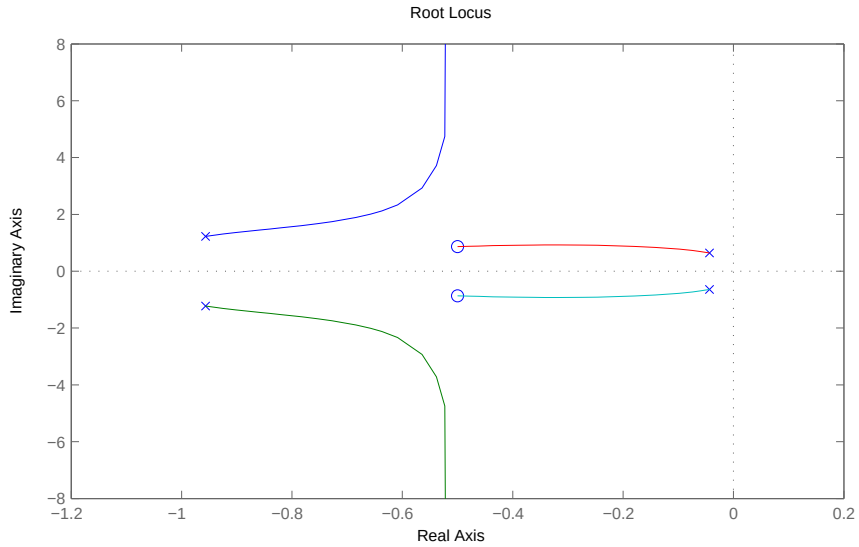
Poles **MAY** destabilize at large gain. But will they???

- Why these poles?

The Asymptotic Center

$$G(s) = \frac{(s - z_1) \cdots (s - z_m)}{(s - p_1) \cdots (s - p_n)}$$

- $m = \#$ of OL zeroes
- $n = \#$ of OL poles



Definition 1.

The **Center of Asymptotes** is where all asymptotes meet.

$$center = \frac{\sum q_{i_{BIG}}}{\#_{i_{BIG}}} = \frac{\sum_i p_i - \sum_i z_i}{n - m}$$

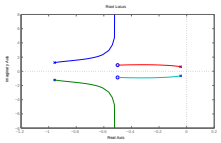
The center of asymptotes is only for the *big* poles on the root locus.

- The center of asymptotes is the average of these points as $k \to \infty$.

The Asymptotic Center

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- $n = \#$ of OL poles



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The center of asymptotes is only for the *big* poles on the root locus.

- The center of asymptotes is the average of these points as $k \to \infty$.

Why? Because when $n-m \geq 2$, the sum of the closed-loop poles doesn't change under feedback.

- Some poles go zeros (z_i)
- Other poles form the asymptotes ($q_{i_{BIG}}$)

Thus

$$\sum_i p_i = \sum_i z_i + \sum_i q_{i_{BIG}}$$

Equivalently

$$\sum_i q_{i_{BIG}} = \sum_i p_i - \sum_i z_i$$

Calculating the Asymptotic Center

Using poles and zeros directly

Expand the formula for asymptotic center:

$$center = \frac{\sum p_i - \sum z_i}{n - m}$$

Typically, we know the p_i and z_i

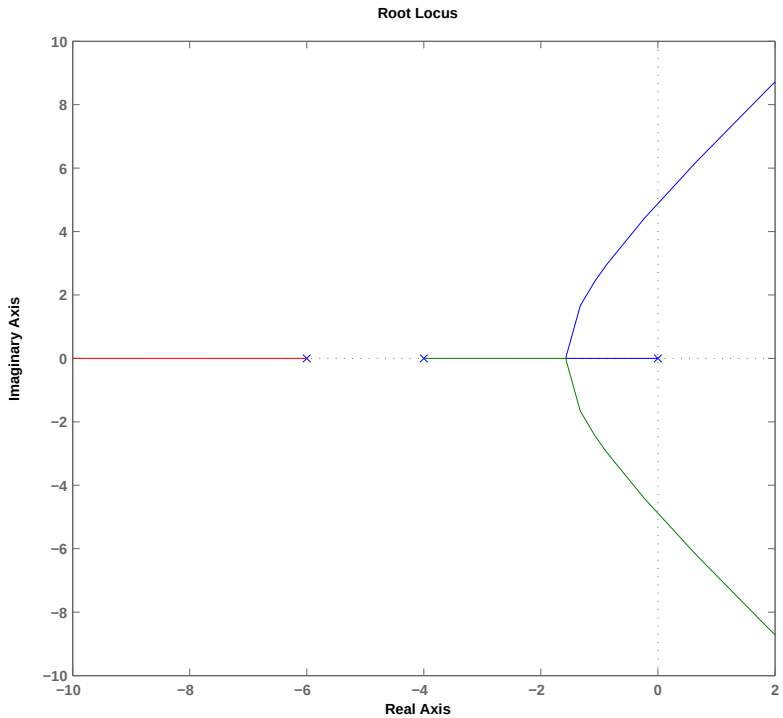
$$G(s) = \frac{1}{s(s+4)(s+6)}$$

First, $\#_{i_{BIG}} = n - m = 3$

Applying the formula:

- No Zeroes
- $p_1 = 0$, $p_2 = -4$, $p_3 = -6$.

$$\sum p_i = -4 - 6 = -10$$



$$center = \frac{\sum p_i - \sum z_i}{n - m} = \frac{-10 - 0}{3} = -3.33\bar{3}$$

A Trick for Calculating the Asymptotic Center

If we just have the transfer function in the form

$$G(s) = \frac{n(s)}{d(s)} = \frac{(s-z_1)\cdots(s-z_m)}{(s-p_1)\cdots(s-p_n)} = \frac{s^m + b_1 s^{m-1} + \cdots + b_m}{s^n + a_1 s^{n-1} + \cdots + a_n}$$

Then

$$\sum z_i = -b_1 \qquad \text{and} \qquad \sum p_i = -a_1$$

Finally (assuming $n - m \geq 2$)

$$center = \frac{\sum q_{i_{BIG}}}{\#_{i_{BIG}}} = \frac{\sum p_i - \sum z_i}{n-m} = \frac{b_1 - a_1}{n-m}$$

Where

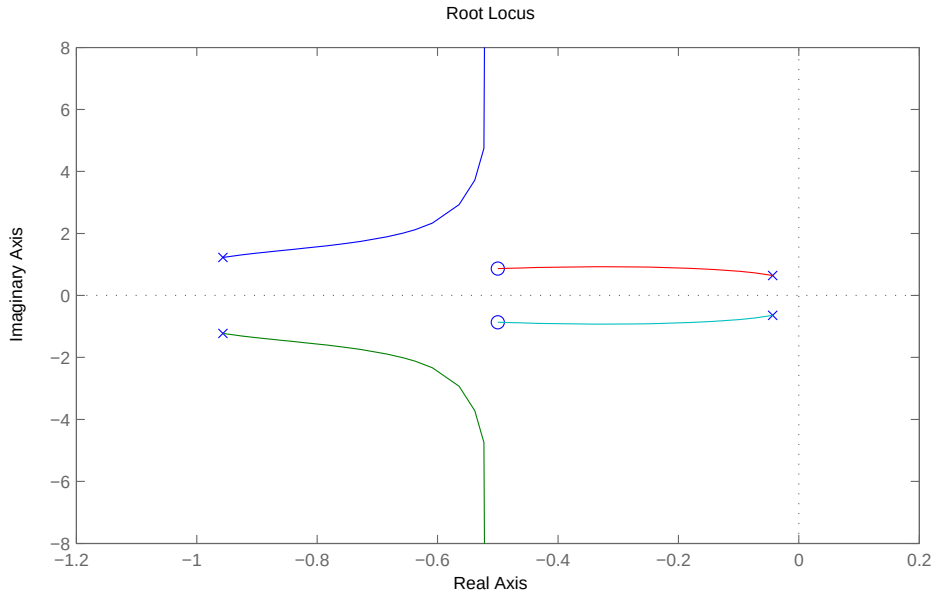
- a_1 is the first coefficient of $d(s)$
- b_1 is the first coefficient of $n(s)$

Calculating the Asymptotic Center

Example: Suspension System

$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + 1s + 1}$$

$$\begin{aligned} \#_{i_{BIG}} &= n - m \\ &= \#poles - \#zeroes \\ &= 2. \end{aligned}$$



Read off the coefficients

- $a_1 = 2$
- $b_1 = 1$

$$center = \frac{b_1 - a_1}{n - m} = \frac{1 - 2}{2} = -\frac{1}{2}$$

Conclusion: High gain is stable.

Calculating the Asymptotic Center

Example: Tweaked Suspension System

Look what happens if we change 2nd coefficient in $n(s)$ from 1 to 3.

$$G(s) = \frac{s^2 + 3s + 1}{s^4 + 2s^3 + 3s^2 + 1s + 1}$$

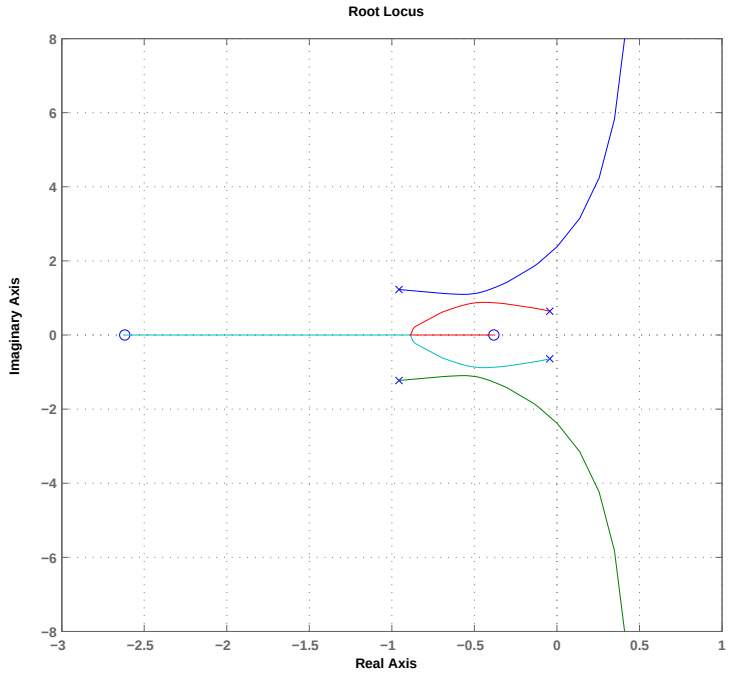
$$\#_{i_{BIG}} = n - m = \#poles - \#zeroes = 2$$

Read off the coefficients

- $a_1 = 2$
- $b_1 = 3$

Thus

$$center = \frac{b_1 - a_1}{n - m} = \frac{3 - 2}{2} = \frac{1}{2}$$



Now high gain is unstable.

Calculating the Asymptotic Center

Example: Suspension System with Integral Feedback

$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + 1s + 1} \frac{1}{s}$$
$$= \frac{s^2 + s + 1}{s^5 + 2s^4 + 3s^3 + 1s^2 + s}$$

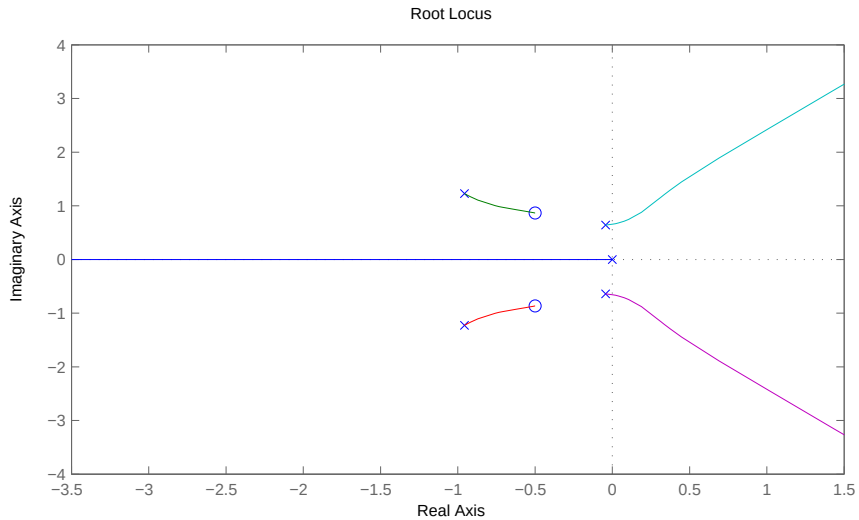
$\#_{i_{BIG}} = n - m =$
$\#poles - \#zeroes = 3.$

Again, we have the same coefficients

- $a_1 = 2$
- $b_1 = 1$

Thus

$$center = \frac{b_1 - a_1}{n - m} = \frac{1 - 2}{3} = -\frac{1}{3}$$



Calculating the Asymptotic Center

Another Example

$$G(s) = \frac{s^2 + s + 1}{s^6 + 2s^5 + 5s^4 - s^3 + 2s^2 + 1}$$

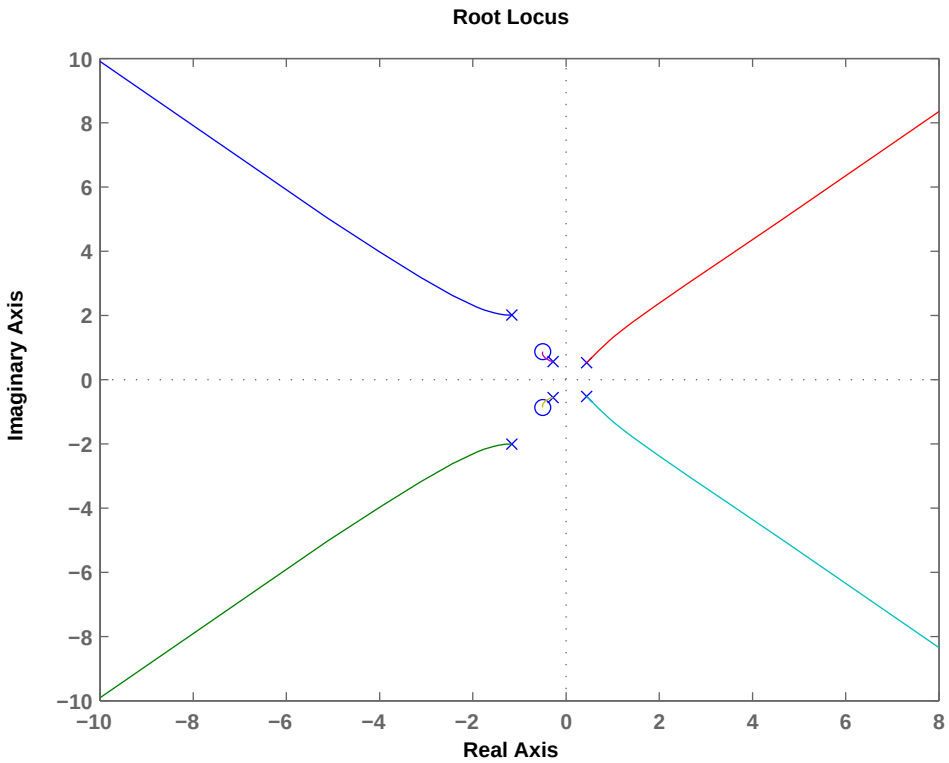
First, $\#_{i_{BIG}} = n - m = 4$.

Again, we have the same coefficients

- $a_1 = 2$
- $b_1 = 1$

Thus

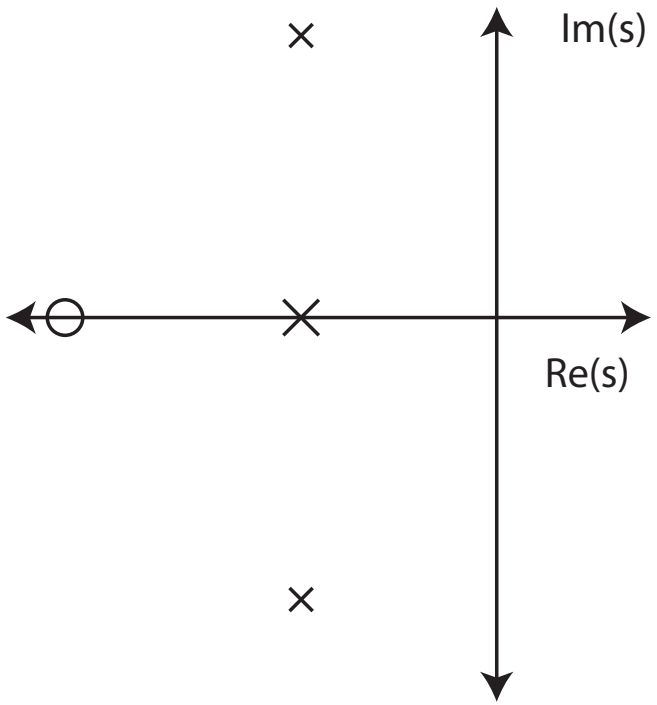
$$center = \frac{b_1 - a_1}{n - m} = \frac{1 - 2}{4} = -\frac{1}{4}$$



Calculating the Asymptotic Center

DIY Example

$$G(s) = \frac{s+2}{(s+1)(s^2+2s+2)}$$



Where does the Root Locus Leave the Real Axis?

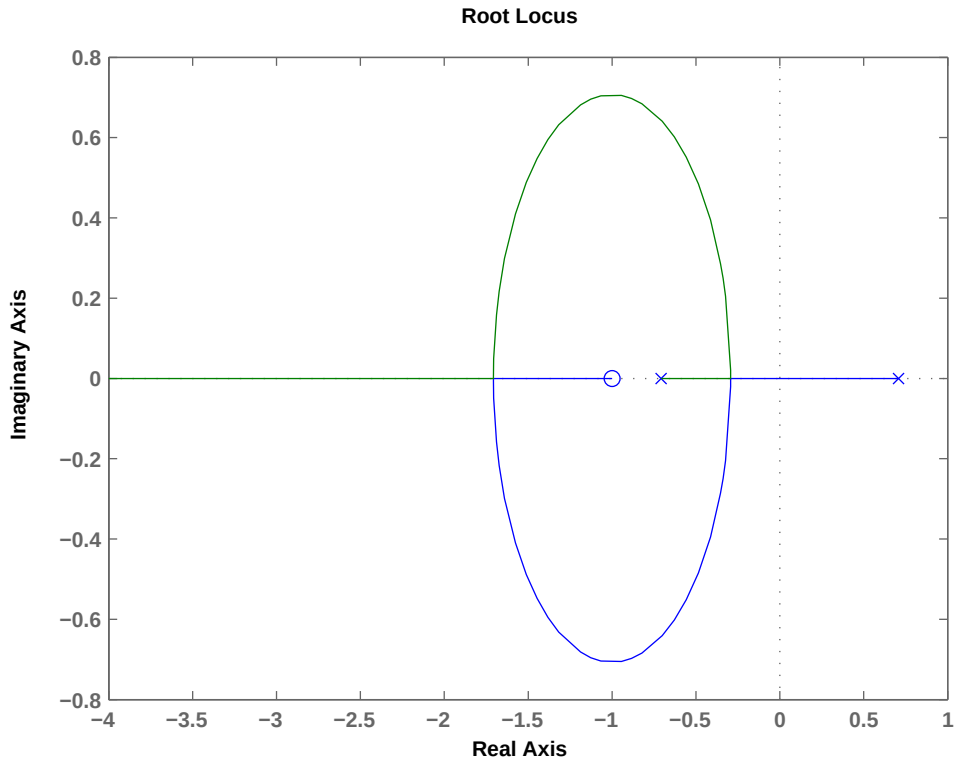
Break points

Recall the inverted pendulum with derivative feedback.

$$G(s) = \frac{1+s}{s^2 - \frac{1}{2}}$$

When do the poles become imaginary?

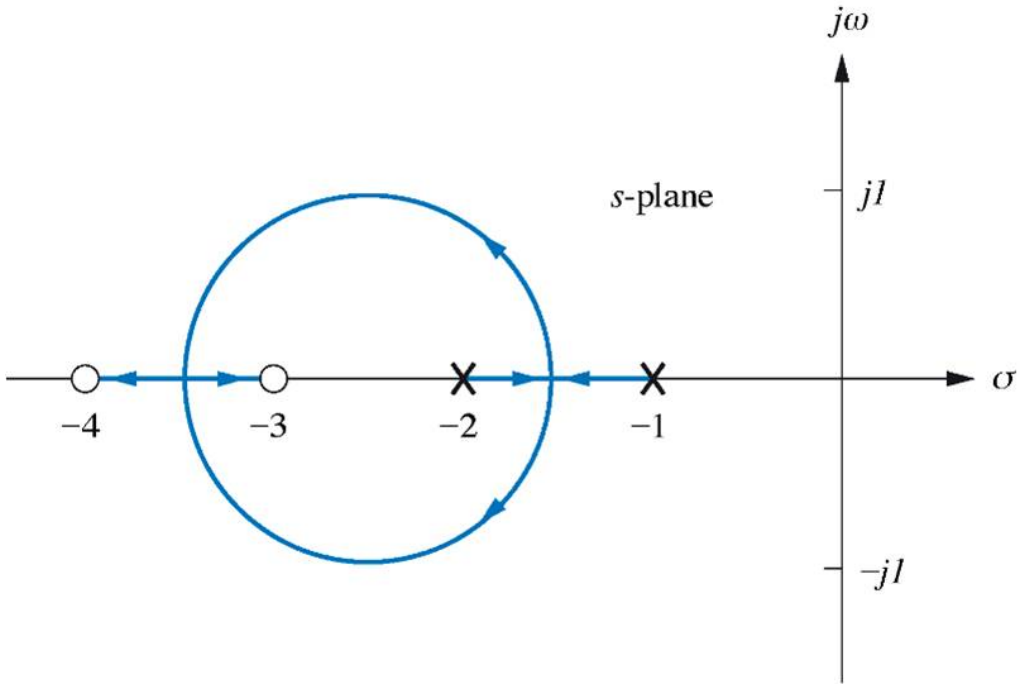
- Important for choosing optimal k.



Break points

Other Examples

$$G(s) = \frac{(s+3)(s+4)}{(s+1)(s+2)}$$



Break points: Maxima and Minima Interpretation

Recall for a point on the root locus

$$d(s) + kn(s) = 0$$

or for a point on the real axis: $s = a$

$$k(a) = -\frac{d(a)}{n(a)} = -\frac{1}{G(a)}$$

Definition 2.

Extrema of a continuous function of a real variable, $k(a)$, occur at

$$\frac{d}{da}k(a) = 0$$

$$\frac{d}{da}k(a) = -\frac{d}{da}\frac{1}{G(a)} = \frac{d(a)}{n(a)^2}n'(a) - \frac{d'(a)}{n(a)} = \frac{d(a)n'(a) - d'(a)n(a)}{n(a)^2} = 0$$

Break Points occur at real-valued solutions of

$$d(a)n'(a) - d'(a)n(a) = 0$$

Break points: Maxima and Minima Interpretation

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Break Points occur at real-valued solutions of

$$d(a)n'(a) - d'(a)n(a) = 0$$

Extrema can also occur at the boundaries, but since we are searching over the whole real axis, there are no boundaries!

Note 1: Real Roots of

$$d(a)n'(a) - d'(a)n(a) = 0$$

which are NOT on the root locus should be ignored. These are NOT break points for the positive root locus.

Note 2: All Complex roots of

$$d(a)n'(a) - d'(a)n(a) = 0$$

should be ignored. These are NOT break points.

Break points

Numerical Example

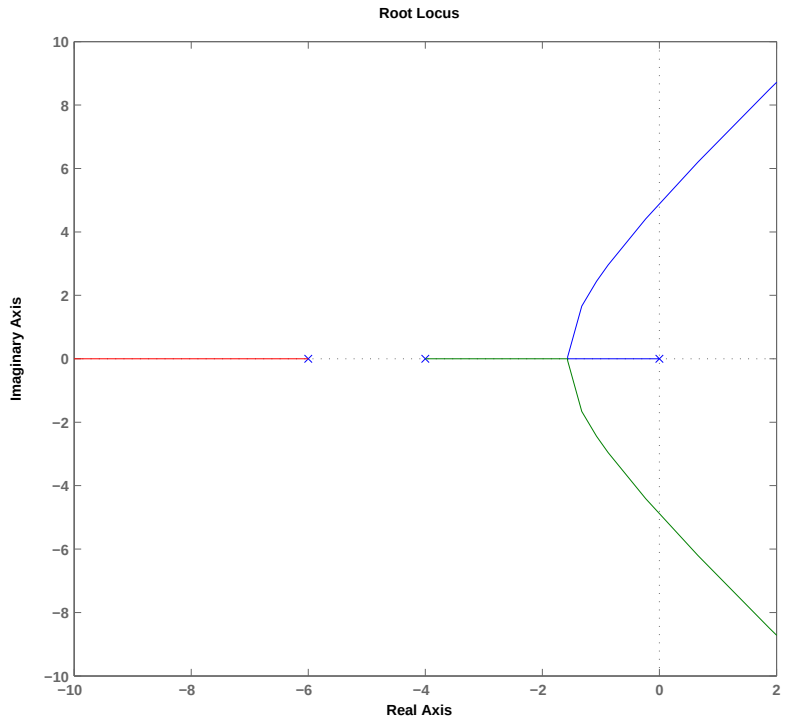
$$G(s) = \frac{1}{s(s+4)(s+6)} = \frac{1}{s^3 + 10s^2 + 24s}$$

Break points occur when

$$\begin{aligned} & d(a)n'(a) - d'(a)n(a) \\ & = 0 - (3a^2 + 20a + 24) = 0 \end{aligned}$$

which has roots

$$\begin{aligned} a_{1,2} &= \frac{-20 \pm \sqrt{20^2 - 4 * 24 * 3}}{6} \\ &\cong -5.1, \, -1.57 \end{aligned}$$



Note: The break point at $a = -5.1$ is not on the root locus and should be ignored!

Break points

Numerical Example

$$G(s) = \frac{(s+3)(s+4)}{(s+1)(s+2)} = \frac{s^2+7s+12}{s^2+3s+2}$$

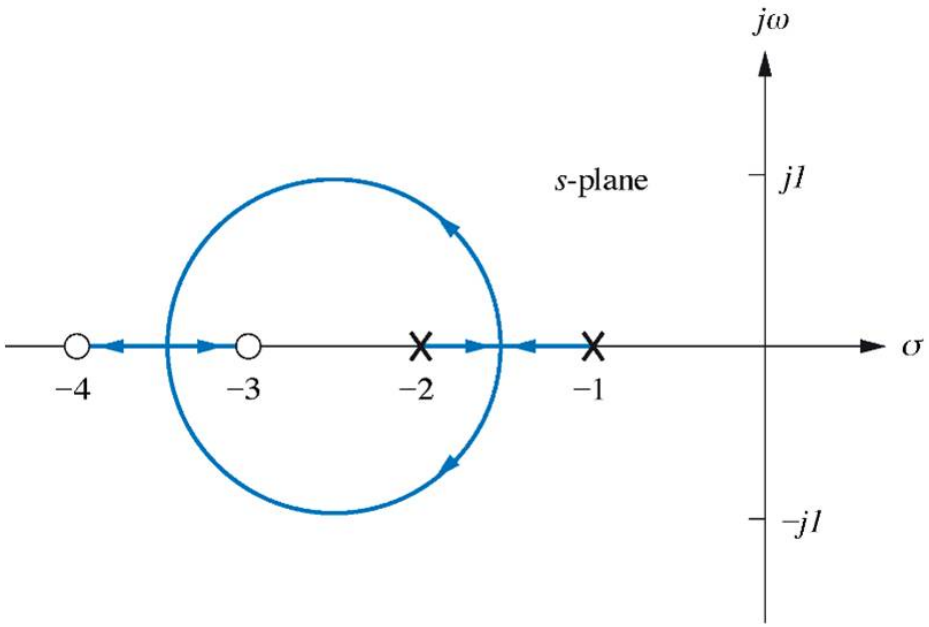
Break points occur when

$$\begin{aligned} d(a)n'(a) - d'(a)n(a) &= (a^2+3a+2)(2a+7) - (2a+3)(a^2+7a+12) \\ &= (a^2+3a+2)(2a+7) - (2a+3)(a^2+7a+12) \\ &= -2(2a^2+10a+11) = 0 \end{aligned}$$

Which has roots

$$a_{1,2} = -1.634,\ -3.366$$

Break points at -1.634 and -3.366.



Break points

Numerical Example

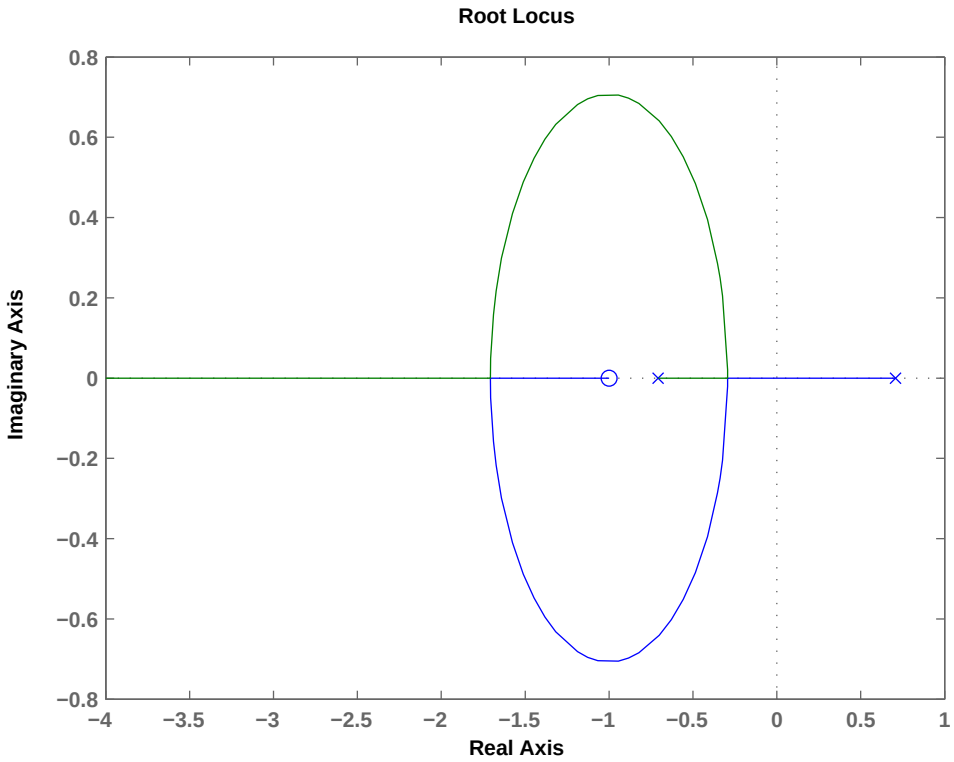
$$G(s) = \frac{1+s}{s^2 - \frac{1}{2}}$$

Break points occur when

$$\begin{aligned} & d(a)n'(a) - d'(a)n(a) \\ & \quad = (a^2 - .5) \cdot 1 - 2a \cdot (1+a) \\ & \quad = -(a^2 + 2a + .5) = 0 \end{aligned}$$

Which has roots

$$a_{1,2} = -.293, \; -1.707$$



Break points at $-.293$ and -1.707

Summary of Root Locus

Step 1: Root Locus starts at Open Loop Poles.
Step 2: At Large Gain, $k \to \infty$

- Small Poles go to zeroes
- Large Poles approach asymptotes
- Center at

$$\sigma = \frac{\sum p_i - \sum z_i}{n-m} = \frac{b_1 - a_1}{n-m}$$

Step 3: On real axis

- When odd number of poles/zeroes to the right.
- Break points when

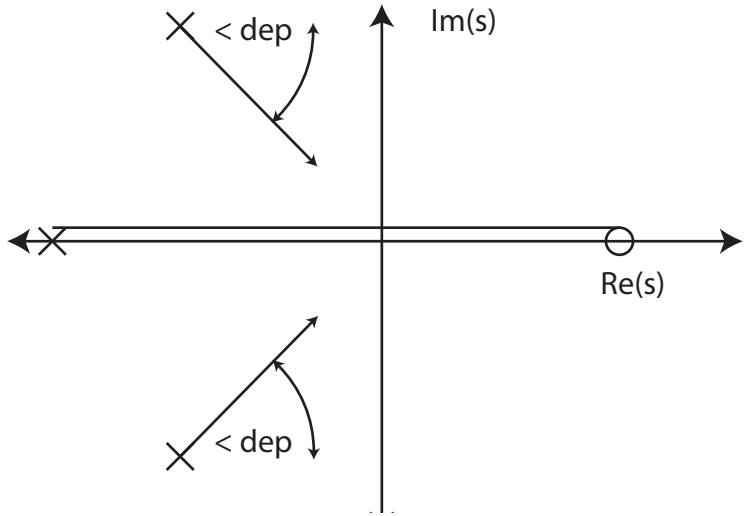
$$-\frac{d}{da}\frac{1}{G(a)} = 0 \qquad \textbf{or} \qquad d(a)n'(a) - d'(a)n(a) = 0$$

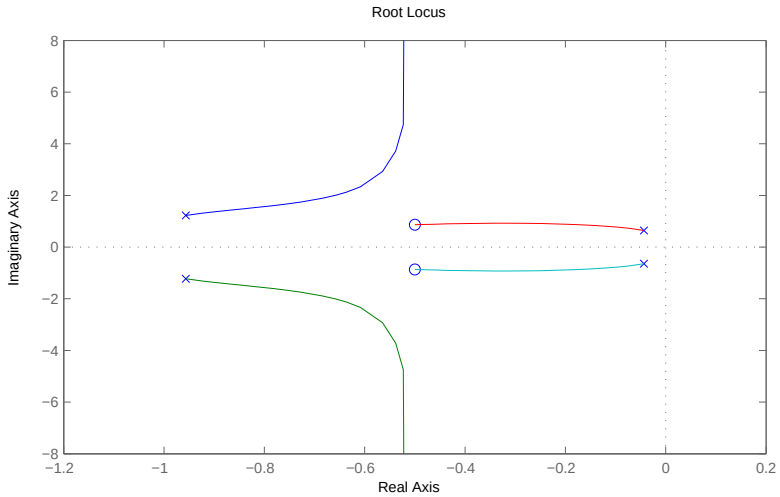
Departure Angles for Complex Poles

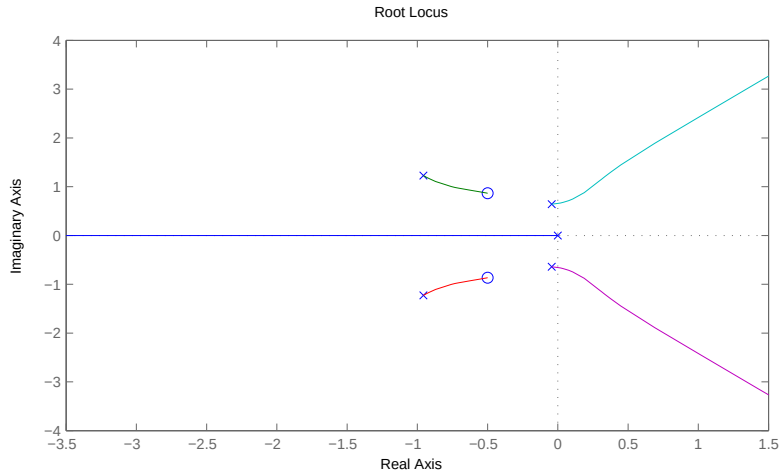
Movement at small gain

The root locus starts at the poles.

- What is the effect of small gain?
- Do the poles become more or less stable?



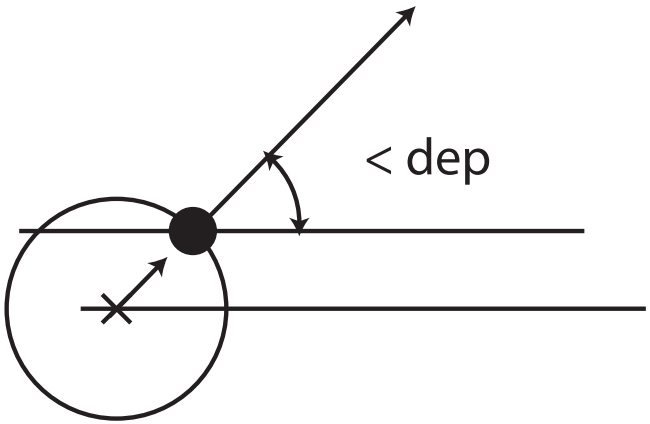


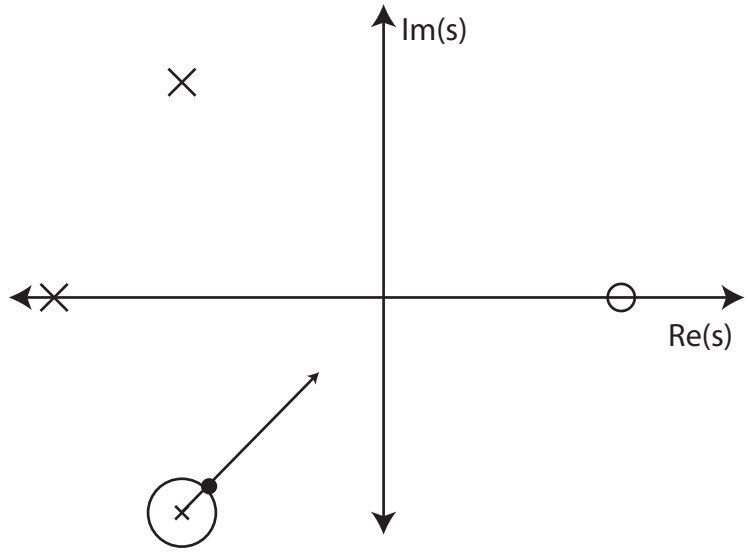


Departure Angles for Complex Poles

Root locus near the open-loop poles

To find the departure angle, we look at a very small region around the departure point.





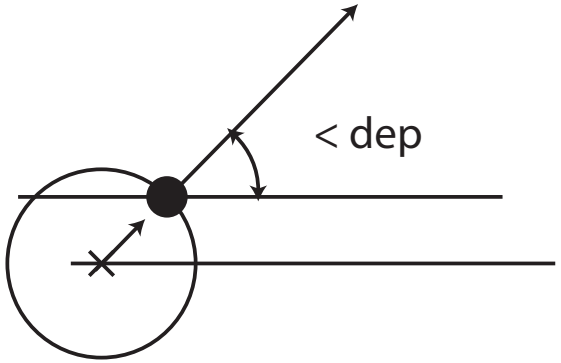
For a point to be on the root locus, we want phase of $180°$.

$$\angle G(s) = \sum_{i=1}^{m} \angle(s - z_i) - \sum_{i=1}^{n} \angle(s - p_i) = 180°$$

Formula for Departure Angle

If we make the point s extremely close to the pole p.

- The angle to other poles and zeros from s is the same as from p.
  - $\angle(s - z_i) \cong \angle(p - z_i)$ for all i
  - $\angle(s - p_i) \cong \angle(p - p_i)$ for all i
- The only difference is the phase from p itself.

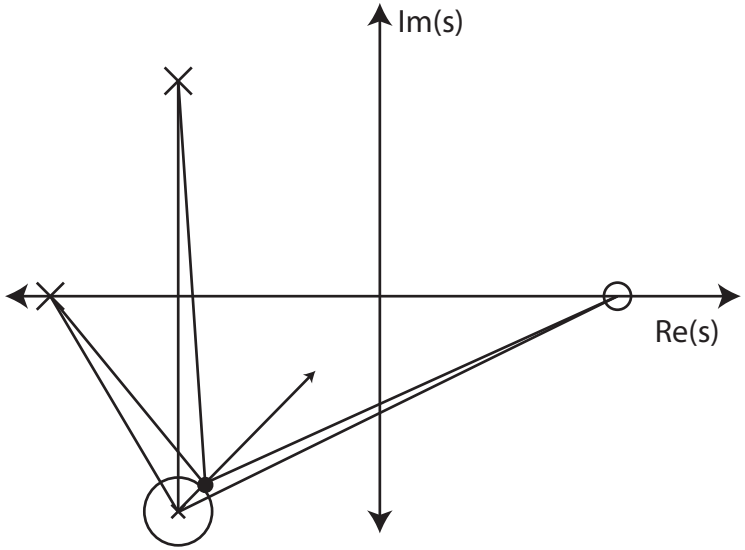


The phase due to p equals the departure angle itself ($\angle_{dep}$)

$$\angle(s - p) = \angle_{dep}$$

The total phase is

$$\angle G(s) \cong \angle G(p) - \angle(s - p) = \angle G(p) - \angle_{dep} = 180^\circ$$



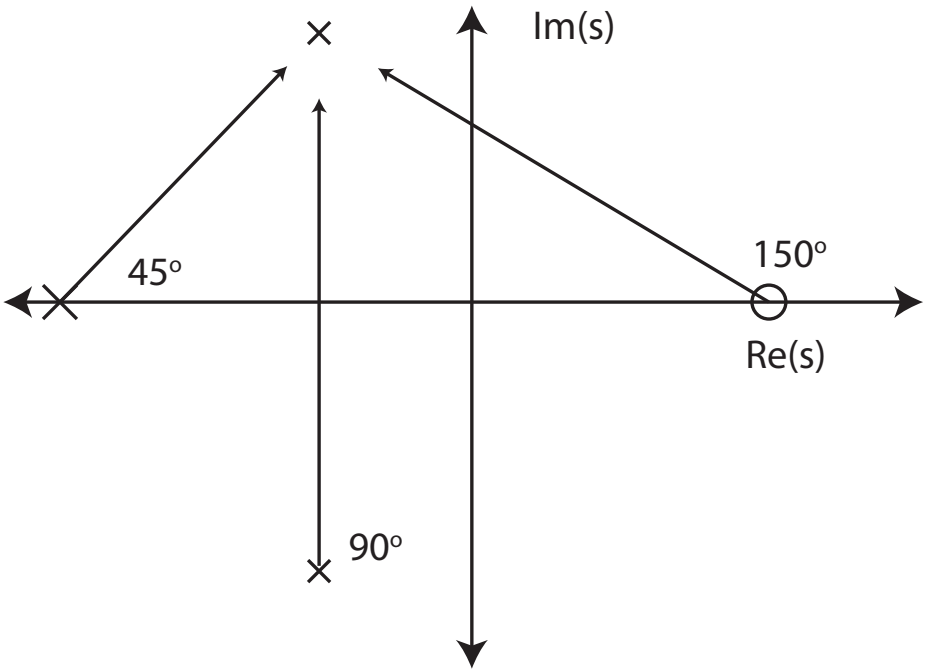
Thus the departure angle from pole p is

$$\angle_{dep} = \angle G(p) + 180^\circ$$

Therefore, to find the departure angle from pole p, just find the phase at p.

Departure Angle

Numerical Example given the angles



The phase at p is based on geometry.

$$\angle G(p) = 150^\circ - 90^\circ - 45^\circ = 15^\circ$$

So the departure angle is easy to calculate.

$$\angle_{dep} = \angle G(p) + 180^\circ = 195^\circ$$

Departure Angle

Numerical Example with complicated angles

$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + 1s + 1}$$

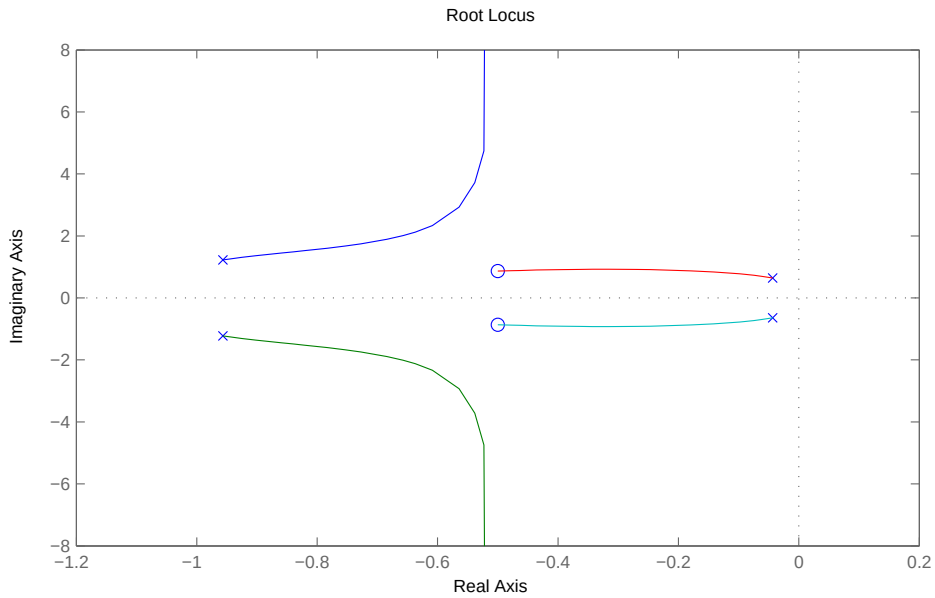
Poles at

- $p_{1,2} = -.957 \pm 1.23\imath$
- $p_{3,4} = -.0433 \pm .641\imath$

Zeroes at

- $z_{1,2} = -.5 \pm .866\imath$

Problem:
Find departure angle at
$p_1 = -.957 + 1.23\imath$.



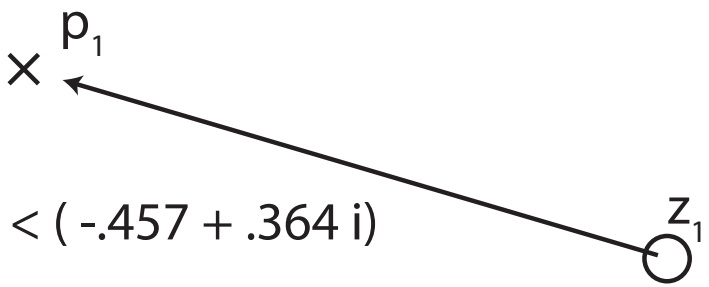
$$\angle_{dep} = 180^\circ + \angle(p_1 - z_1) + \angle(p_1 - z_2) - \angle(p_1 - p_2) - \angle(p_1 - p_3) - \angle(p_1 - p_4)$$

The difficulty is calculating the phase.

Departure Angle

Numerical Examples

$$\begin{aligned}\angle(p_1 - z_1) &= \angle(-.957 + 1.23\imath + .5 - .866\imath) \\ &= \angle(-.457 + .364\imath) \\ &= \tan^{-1}\left(\frac{.364}{-.457}\right) \\ &= 141.46^\circ\end{aligned}$$



$$\angle(p_1 - z_2) = \angle(-.457 + 2.096\imath) = 102.3^\circ$$

Obviously,

$$\angle(p_1 - p_2) = 90^\circ$$

$$\angle(p_1 - p_3) = 147.2^\circ, \qquad \angle(p_1 - p_4) = 116.03^\circ$$

Departure Angle

Numerical Examples

Now that we have all the angles:

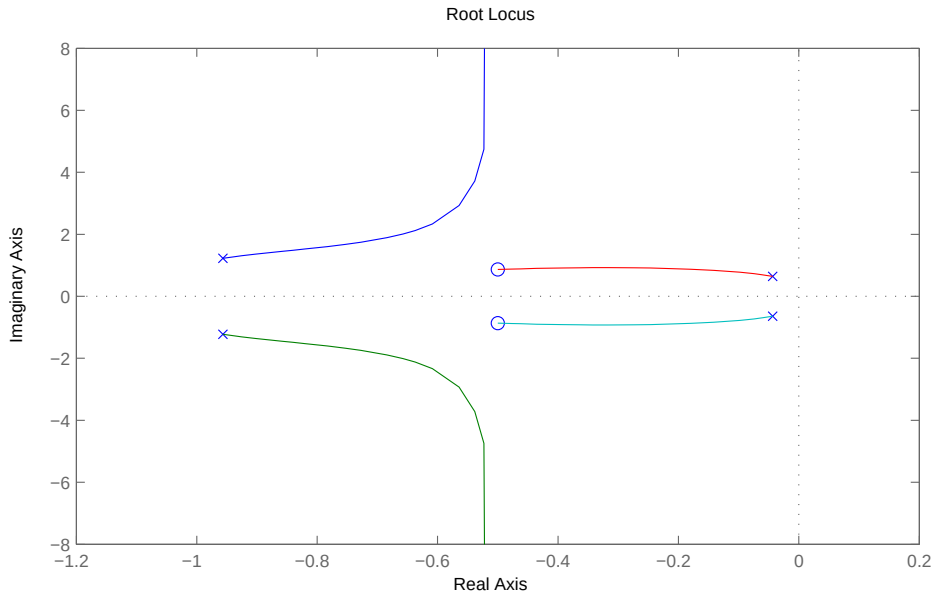
$$\begin{aligned}\angle G(p_1) &= \angle(p_1 - z_1) + \angle(p_1 - z_2) - \angle(p_1 - p_2) - \angle(p_1 - p_3) - \angle(p_1 - p_4) \\ &= 141.46^\circ + 102.3^\circ - 90^\circ - 147.2^\circ - 116.03^\circ \\ &= -109.47^\circ\end{aligned}$$

We conclude

$$\angle_{dep,p_1} = \angle G(p_1) + 180^\circ = 70.53^\circ$$

By symmetry we could find

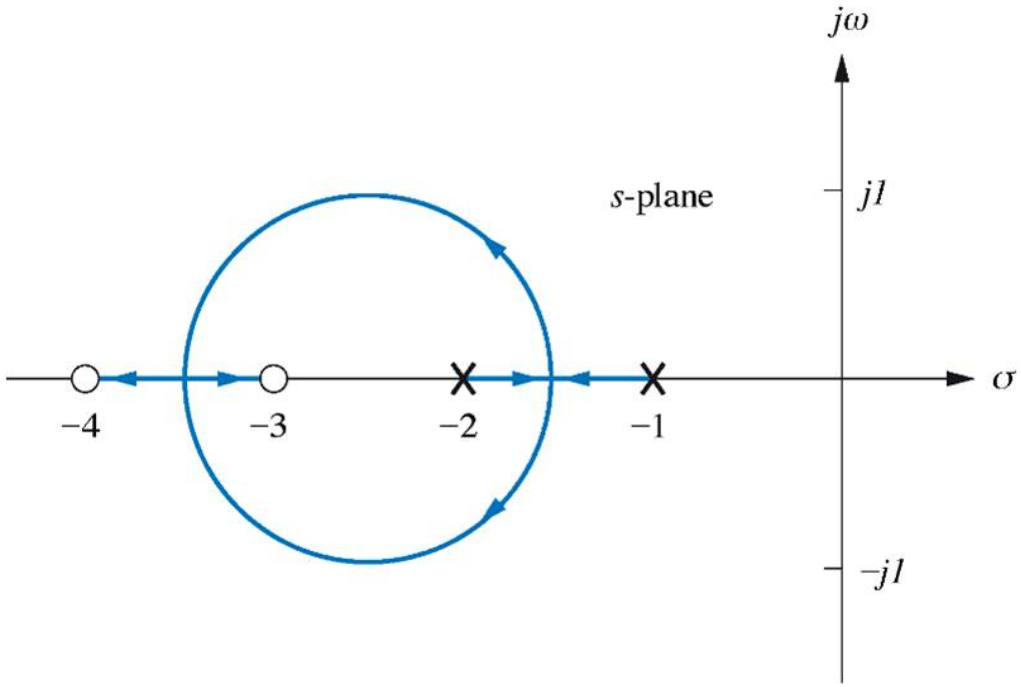
$$\angle_{dep,p_2} = -70.53^\circ$$



Departure Angle

Numerical Examples

What about a pole on the real axis?

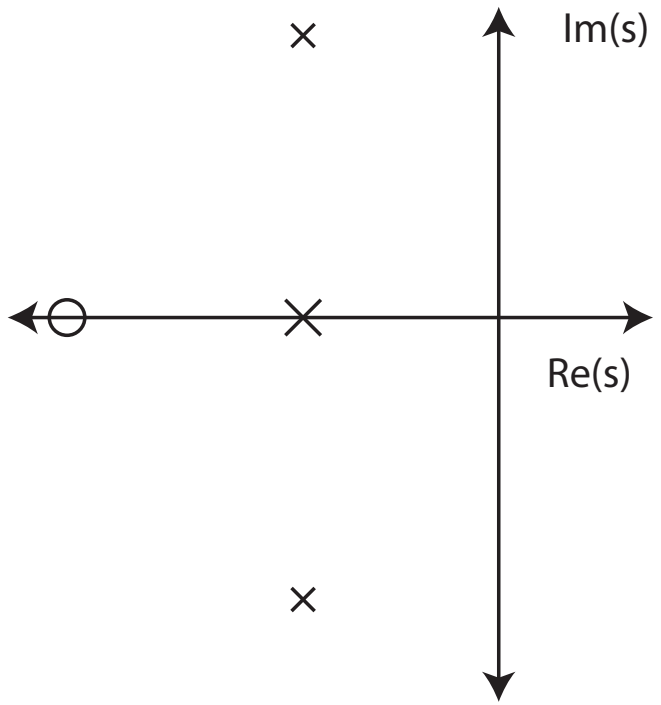


$$\angle G(p) = 0^\circ \quad \text{or} \quad 180^\circ$$

Calculating the Departure Angle

DIY Example

$$G(s) = \frac{s+2}{(s+1)(s^2+2s+2)}$$



Calculating Arrival Angles at Complex Zeros

Behaviour at large gain

To determine where the poles go, compute **Arrival Angles** at zero, z.

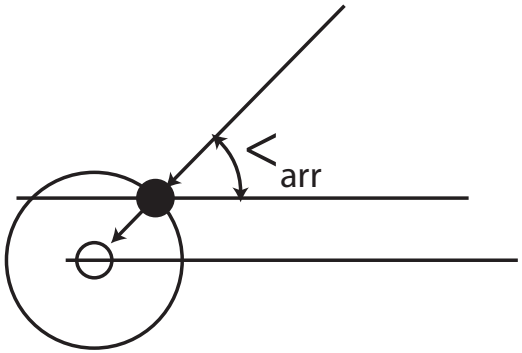
- Very similar to departure angles.

First, draw a small circle around z.

- Calculate the phase at z – i.e. $\angle G(z)$

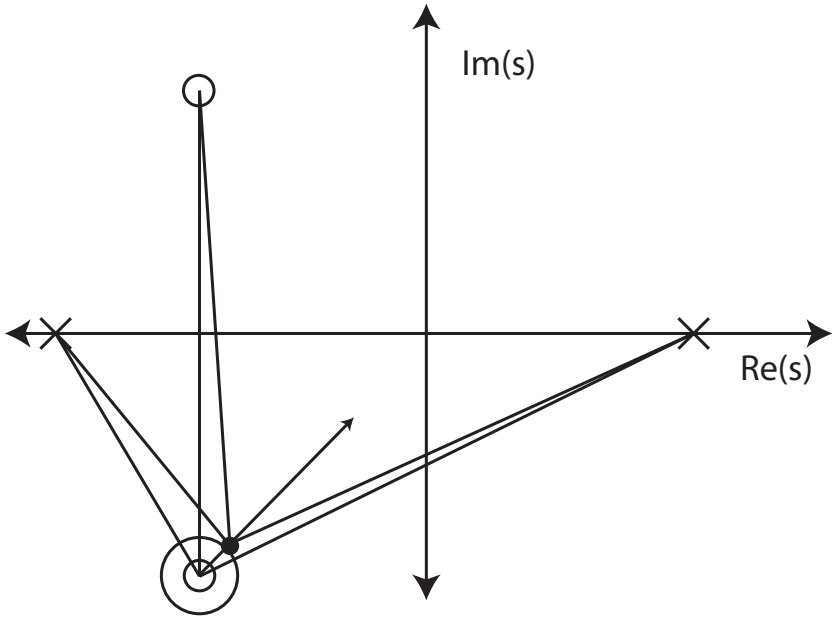
The phase at a point, s, on the circle is.

$$\begin{aligned}\angle G(s) &= \sum_{i=1}^{m} \angle(s - z_i) - \sum_{i=1}^{n} \angle(s - p_i) \\ &= \angle G(z) + \angle_{arr} = 180^\circ\end{aligned}$$



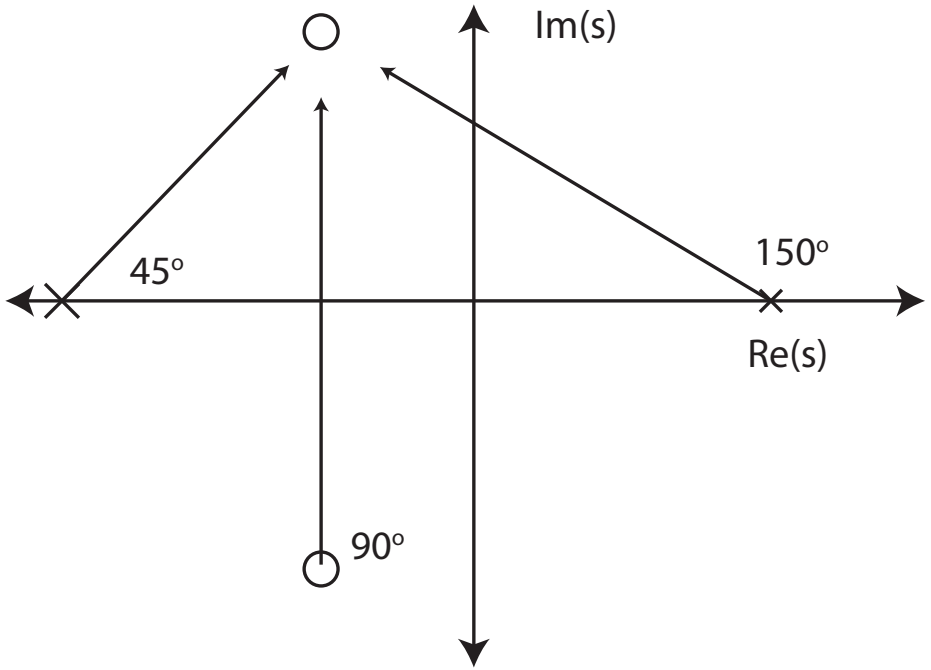
Thus the arrival angle is

$$\angle_{arr} = 180^\circ - \angle G(z)$$



Computing the Arrival Angle

Numerical Example



The phase at z is based on geometry.

$$\angle G(z) = 90^\circ - 150^\circ - 45^\circ = -105^\circ$$

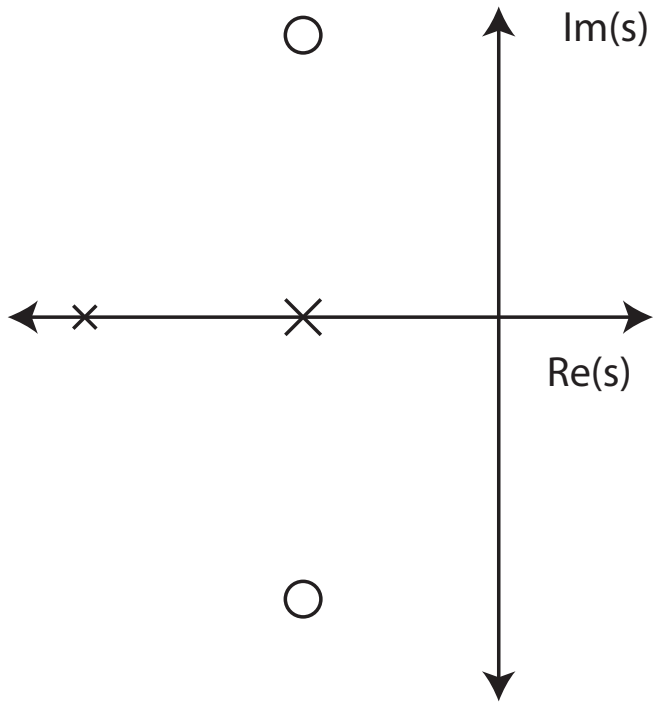
So the arrival angle is easy to calculate.

$$\angle_{arr} = 180^\circ - \angle G(z) = 285^\circ$$

Calculating the Arrival Angle

DIY Example

$$G(s) = \frac{s^2 + 2s + 2}{(s+1)(s+2)}$$



Summary

What have we learned today?

Asymptotic Center

- Where do the asymptotes meet?

Where does the Root Locus Leave the Real Axis

- Break Points

What is the effect of small gain?

- Departure Angles from Complex Poles

What is the behaviour at high gain?

- Arrival Angles at Complex Zeroes

Next Lecture: Computing Gain, Generalized Root Locus